

System Architecture Guide

Component Relationships, Database schemas, and Data Analytics Flow

One Janitorial Operations Library

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1. Overview & Purpose

Architectural blueprint of the platform, outlining server tiers, database layouts, and network components.

Helps developers understand the data relationships and system dependencies.

Details relational structures mapping to Datasets, Analytics Dashboards, and logs.

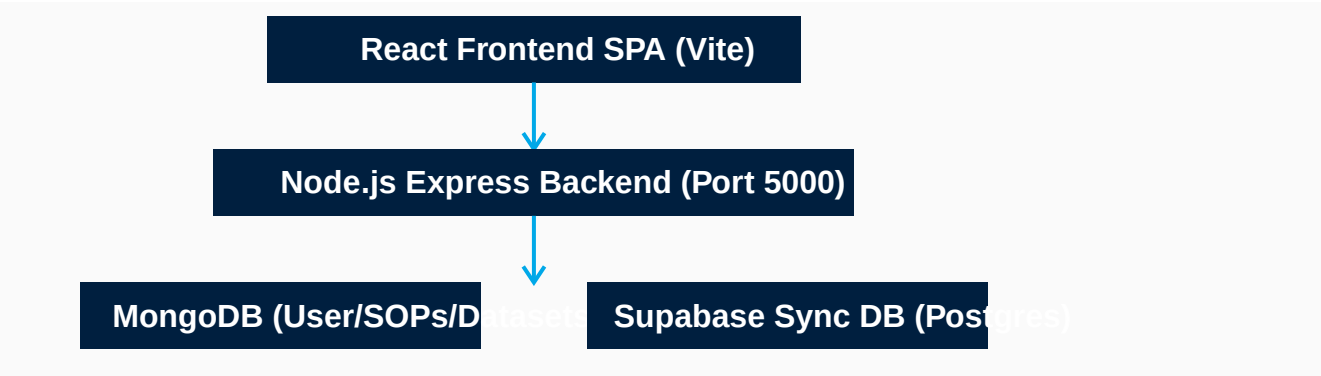
2. Key Features & Blueprint

Three-Tier Architecture: React frontend client, Node.js API server, and MongoDB/PostgreSQL database layers.

Data Analytics Flow: Pipeline processing inputs, executing schema detection, and caching outputs.

Task Queue Framework: Redis and BullMQ processes handling asynchronous actions.

3. System Process Flow



4. Step-by-Step Usage & Operations

Reviewing stack components: Inspect docker compose configurations to identify server structures.

Mapping database schemas: Check mongoose model folders to map dataset and dashboard layouts.

Auditing Data replication: Check Supabase sync logs to verify Postgres schema matches.

5. Best Practices & Safety Guidelines

- Use database caching schemas to speed up repetitive query executions.
- Keep database server instances inside secure private subnets to limit access.
- Verify that dataset upload folder permissions restrict execution execution scripts.

6. Troubleshooting & FAQ

- Slow request responses: Check database query statistics and run query execution plans.
- Upload file storage full: Run cleanup crons to purge historical csv temp files.
- Queue execution lags: Confirm worker threads count and review Redis server status.

7. Glossary & Appendix

- Express: Node.js web frame.
- MongoDB: Primary database.
- Lineage Tracker: Sequence tracking database modifications applied to tables.